SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 9

Capital Budgeting: Evaluation Techniques & Project Cash Flows

Capital budgeting is the process of evaluating and selecting long-term investments that increase owners' wealth. This week uses one running case (Bennett Company, Projects A and B) for four tools: payback period, NPV (with the profitability index and EVA) and IRR — then explains why NPV and IRR can rank projects differently (NPV profile, reinvestment assumption, timing of cash flows, size of investment). The second part answers where the cash flows come from: relevant cash flows (incremental, after tax, cash not profit), sunk vs opportunity costs, the initial investment (including tax on selling the old asset and working capital), operating cash flows and the terminal cash flow, using the Powell Corporation case through to the final decision.

Material

1 What capital budgeting is, and its process

Capital budgeting = the process of evaluating and selecting long-term investments that *increase owners' wealth*. It focuses on **capital expenditure**: benefits last more than a year, amounts are large, and decisions are hard to reverse.

Capital expenditure	Operating expenditure
Benefits last > 1 year	Benefits used up within 1 year
Buying a new machine, building a factory/store, launching a new product line	Paying salaries, buying raw materials, monthly utilities and rent

1. Proposal generation (managers at all levels)
→ 2. Review & analysis (finance assesses)
→ 3. Decision (big outlays go to the board)

→ 4. Implementation (money spent in phases)

→ 5. Follow-up (actual vs forecast)

This week's tools sit in **step 2**: they turn a proposal's cash flows into an accept or reject decision. Step 5 is often forgotten, yet it is where the firm learns whether its forecasts were too optimistic.

Three pairs of key terms:

Term	Meaning	Its opposite
Independent projects	Cash flows unrelated; accepting one does not affect the others	**Mutually exclusive**: projects compete; choosing one rules out the others (e.g. two machines for the same job)
Unlimited funds	Every project earning the minimum return can be accepted	**Capital rationing**: a fixed budget; projects compete for money
Accept–reject approach	Does the project meet the minimum standard?	**Ranking approach**: which project is best on a chosen measure?

A good technique
accounts for the *time value of money* (a dollar today beats a dollar later) and the *risk–return trade-off*. On this test, payback is the weakest tool and NPV the strongest.

2 Running case: Bennett Company, Projects A and B

Bennett, a medium-sized metal fabricator, is considering two projects. Every technique this week is applied to these same cash flows:

End of year	0	1	2	3	4	5
Project A	–$420,000	$140,000	$140,000	$140,000	$140,000	$140,000

Project B	– $450,000	$280,000	$120,000	$100,000	$100,000	$100,000

Project A pays the same amount each year (an **annuity**). Project B pays a lot early and less later (a **mixed stream**). Bennett's cost of capital is 10% and its maximum acceptable payback is 2.75 years. This difference in timing is what later makes NPV and IRR rank the projects differently.

3 Payback period: how fast does the money come back?

The **payback period** = the time it takes a project's cash inflows to recover the initial investment.

```
Equal cash flows: Payback = Initial investment ÷ Annual cash
inflow
Unequal cash flows: add up inflows year by year until the
investment is recovered
```

Rule: accept if payback < the maximum set by management; reject if longer. Shorter is better: the money comes back sooner and is exposed to risk for less time.

≡ **Answer to Example 10.1 (Bennett, maximum 2.75 years)**
Project A: 420,000 ÷ 140,000 = **3.0 years** → rejected (3.0 > 2.75).
Project B: years 1 + 2 = 280,000 + 120,000 = 400,000; the remaining 50,000 is recovered in year 3 (50,000 ÷ 100,000 = 0.5) → **2.5 years** → accepted.
If only one can be chosen, payback prefers **B**.

Strengths

Simple and intuitive; widely used (especially by small firms and for small projects); gives some weight to timing (faster recovery is rewarded); a crude risk check — riskier projects can be required to pay back faster.

vs

Weaknesses

The cut-off is chosen subjectively; no direct link to maximising shareholder wealth; ignores cash flows after payback; does not properly discount for the time value of money.

An example of the weakness: one project pays back in 2 years and then stops; another pays back in 3 years and keeps earning for 10 more — payback picks the first, though the second is worth far more.

4 Net present value (NPV): does the project create value?

A dollar today is worth more than a dollar later because it can be invested. $100 today at 10% becomes $110 in a year; equally, $110 in a year is worth $100 today. **Discounting** converts future cash flows into today's value (PV). The discount rate r = the firm's **cost of capital** (the return investors require — see WACC in week 8).

$$NPV = \Sigma\ CF_t \div (1 + r)^t - CF_0$$

CF_t = cash inflow in year t; CF_0 = initial investment; r = cost of capital. **Rule:** accept if NPV > 0; reject if NPV < 0; to rank, higher NPV is better.

Why it works: a positive NPV means the project earns more than investors require, so it adds that amount to the owners' wealth.

≡ **Answer to Example 10.5 (cost of capital 10%)**

Project A (annuity): PV = 140,000 × [1 – 1.1^{-5}] ÷ 0.1 = 140,000 × 3.7908 = 530,710 → NPV = 530,710 – 420,000 = **$110,710**.

Project B (mixed): 280,000/1.1 + 120,000/1.1^2 + 100,000/1.1^3 + 100,000/1.1^4 + 100,000/1.1^5 = 254,545 + 99,174 + 75,131 + 68,301 + 62,092 = 559,244 → NPV = 559,244 – 450,000 = **$109,244**.

Both are acceptable (NPV > 0). If only one: **A** (higher NPV) — the opposite of payback's choice.

In Excel: `=NPV(10%, CF1:CF5) + CF0`. Careful: Excel's NPV assumes the first cash flow arrives at the end of year 1, so CF0 is added outside the function.

5 Profitability index (PI) and Economic Value Added (EVA)

```
PI = PV of cash inflows ÷ Initial investment
```

Rule: accept if PI > 1.0; reject if < 1.0. PI > 1 is the same as NPV > 0, so NPV and PI *always agree* on accept/reject. PI shows value created per $1 invested — useful under **capital rationing**.

Bennett	Project A	Project B
PV of inflows ÷ investment	530,710 ÷ 420,000	559,244 ÷ 450,000
PI	**1.26**	**1.24**

But PI can rank differently from NPV: a small project can have a high PI yet a small NPV (wealth added).

```
EVA = Annual cash flow - (Cost of capital × Capital invested)
```

1. Start like NPV Compute the project's net cash flow each year.

2. Subtract a capital charge The return investors demand on the money invested.

3. Read the result Positive EVA = pure economic profit, above what investors require.

NPV measures value over the whole project life; EVA measures performance **year by year** (useful for judging managers). The present value of all EVAs = NPV.

≡ **Answer to Example 10.7**
A $1,000,000 project, $120,000 a year forever, cost of capital 10%.
NPV = 120,000 ÷ 0.10 – 1,000,000 = 1,200,000 – 1,000,000 = **$200,000**.
Annual EVA = 120,000 – (0.10 × 1,000,000) = **$20,000**.
PV of EVA = 20,000 ÷ 0.10 = $200,000 = NPV ✓ → worth doing.

6 Internal rate of return (IRR)

IRR = the discount rate that makes a project's NPV exactly $0. It is the annual return the project earns.

$$0 = \Sigma\ CF_t \div (1 + IRR)^t - CF_0$$

Rule: accept if IRR > cost of capital; reject if IRR < cost of capital; to rank, higher IRR is better. Find it with Excel `=IRR(CF0:CF5)`, a financial calculator, or trial and error by hand.

Answer to Example 10.8

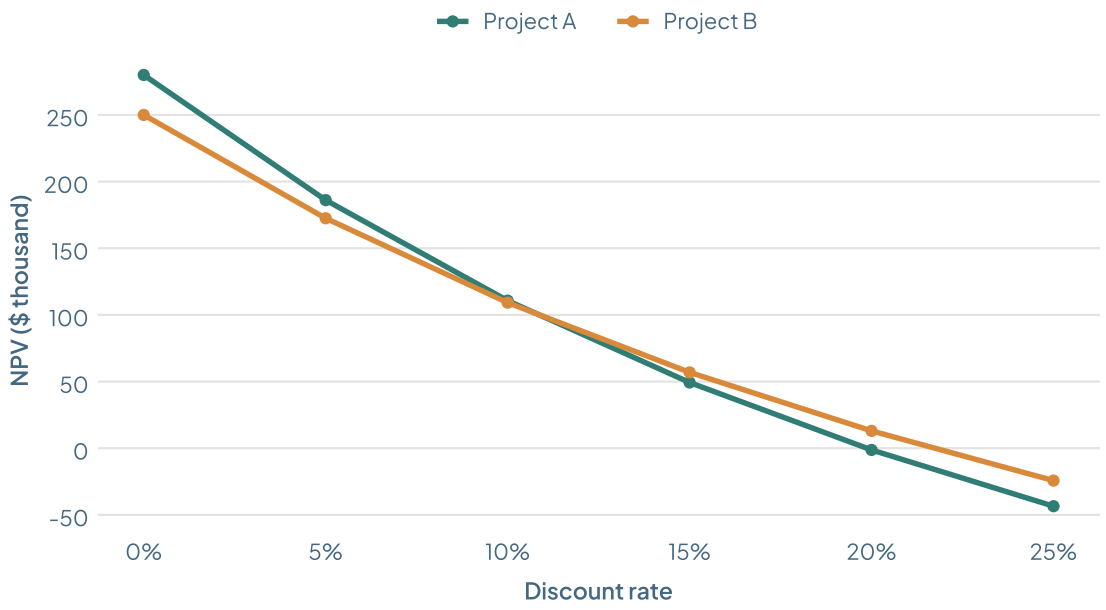
Project A: find r so that 140,000 × the 5–year annuity factor = 420,000 → factor = 3.0 → r ≈ **19.9%**.
Project B: trial and error → r ≈ **21.7%**.
Both > 10% → accepted. IRR ranks **B** first, while NPV ranks **A** first → the methods conflict.

Trial and error: if NPV at rate r is still positive, raise r; if negative, lower it. For A: at 20% NPV = –1,300 (slightly negative), at 19% slightly positive → IRR is just under 20%.

7 NPV vs IRR: the NPV profile and why they can disagree

An **NPV profile** = a graph of a project's NPV at different discount rates. Three easy points: (1) at 0%, NPV = sum of inflows – investment (A: 700,000 – 420,000 = **$280,000**; B: 700,000 – 450,000 = **$250,000**); (2) at the cost of capital = the NPV already computed; (3) at the IRR, NPV = 0.

Bennett NPV profile: Project A vs B



The lines cross near 10.7%. At the 10% cost of capital A's NPV is slightly higher; IRRs are A 19.9% and B 21.7% (where each line crosses zero).

A's line is steeper (its cash flows are level/later, so more sensitive to the discount rate). The lines **cross at about 10.7%**. Below 10.7% A is better; above it B is better. Because Bennett's cost of capital is 10% (below the crossover), NPV picks A, while IRR (which always "looks at" the zero point) picks B.

Cause of conflict	Explanation
Reinvestment assumption	NPV assumes early cash is reinvested at the cost of capital (realistic). IRR assumes it is reinvested at the IRR itself (often too optimistic).
Timing of cash flows	Projects that pay back early (B) are less sensitive to the discount rate than projects paying evenly or late (A).
Size of the investment	IRR is a percentage; it ignores how many dollars are at stake.

≡ **Answer to the size example**

Option 1: invest $2, receive $3 tomorrow → IRR 50%, wealth gained $1. Option 2: invest $1,000, receive $1,100 → IRR 10%, wealth gained $100. IRR prefers option 1, but option 2 makes you **$99 richer**. Choose option 2.

Important: NPV and IRR *always agree on accept/reject*. They can differ only when *ranking* competing projects.

In theory: NPV

Measures exactly how much wealth is created; realistic reinvestment rate; always one answer (IRR can have several when cash flows change sign more than once).

vs

In practice: IRR

Managers like rates of return; easy to compare with the cost of capital. A CFO survey: 76% use IRR, 75% use NPV.

The slides' bottom line

Decide with NPV; use IRR and payback as extra information.

8 Recap of the four techniques

Technique	Question	Accept if	Bennett result	Watch out for
Payback	How fast is the money back?	< maximum period	A 3.0 yrs; B 2.5 yrs	Ignores later cash flows and time value
NPV	How much value is created?	> $0	A $110,710; B $109,244	Needs a good cost-of-capital estimate
PI	Value per $1 invested?	> 1.0	A 1.26; B 1.24	Can rank differently from NPV
EVA	Profit after the cost of capital?	> $0	Its present value = NPV	Year-by-year view
IRR	What return does it earn?	> cost of capital	A 19.9%; B 21.7%	Can conflict with NPV on rankings

9

Calculator: try payback, NPV, PI and IRR yourself

Enter a project's cash flows and the discount rate, and see all four tools at once. Start from the Bennett (A vs B) or Powell case, then change the numbers — e.g. raise r to 15% and watch the NPV ranking of A vs B flip (the NPV profiles cross near 10.7%).

Capital budgeting calculator

Fill with: Bennett A vs B | Powell | Blank

Enter the initial investment as a positive number (treated as cash out), then each year's cash inflow. Leave Project B empty to evaluate one project. Results update as you type.

Cost of capital / discount rate (r): 10 %

Year	Project A	Project B
0 investment	420000	450000
1	140000	280000
2	140000	120000
3	140000	100000
4	140000	100000
5	140000	100000

+ Year | – Year

Project A

NPV	$110,710	accept
IRR	19.9%	accept
PI	1.26	accept
Payback	3.00 years	

▶ Show working

Project B

NPV	$109,244	accept
IRR	21.7%	accept
PI	1.24	accept
Payback	2.50 years	

▶ Show working

If mutually exclusive, choose the higher NPV: **Project A**. IRR picks Project B instead — the rankings conflict, so follow NPV.

10 Relevant cash flows: what counts?

So far Bennett's cash flows were given. In practice the analyst must **estimate them first**, following three rules:

1. Incremental

Count only cash that changes because of the project: "with the project" minus "without it". Replacement example: new machine minus old machine.

2. After tax

Taxes are real cash leaving the firm. Example: selling an old asset above book value triggers tax.

3. Cash, not profit

Accounting profit deducts depreciation, which is not a cash payment. Add it back.

Three types of project cash flow:

Initial investment (year 0): cash out to start

→ Operating cash flows (years 1–n): extra after-tax cash each year

→ Terminal cash flow (final year): after-tax cash from closing down

Expansion project: the new asset's cash flows only. **Replacement** project: the new asset's cash flows minus the old asset's.

Sunk cost → IGNORE

Money already spent that cannot be recovered whatever we decide now. Example: market research paid for last year.

vs

Opportunity cost → INCLUDE

Cash given up by using an asset we already own instead of its best other use. Example: our own empty warehouse that could be rented out.

The test

Does this cash flow change if we accept the project? If yes, it counts.

Answer to Example 11.1 (Jankow Equipment)

The $237,000 paid three years ago for drill press X12 is a **sunk cost** → irrelevant, it is gone whatever is decided. The old machine's $42,000 sale value is an **opportunity cost** → relevant: using its control system for the upgrade means giving up the $42,000 it could be sold for. Include $42,000 as a cash outflow (cost) of the upgrade.

11 Initial investment: asset cost, selling the old asset, and working capital

```
Initial investment = Installed cost of new asset - After-tax
proceeds from selling old asset ± Change in net working capital
```

Installed cost = price + installation (everything needed to make the asset work); this is the amount depreciated. In an expansion project there is no old asset, so the second term = 0.

Selling the old asset — book value and tax. Book value = installed cost – accumulated depreciation. Tax depends on the sale price versus book value:

Sale price (book value $48,000, tax 21%)	Versus book value	Tax effect
$110,000	Gain of $62,000	Pay $13,020 tax
$70,000	Gain of $22,000	Pay $4,620 tax
$48,000	No gain or loss	No tax
$30,000	Loss of $18,000	Save $3,780 tax

Rule: sell above book value → the gain is taxed (cash outflow); below book value → the loss reduces tax (cash inflow). After-tax proceeds = sale price – tax (or + tax saving).

```
Δ Net working capital = Δ Current assets - Δ Current liabilities
```

At the start: extra inventory, receivables and cash are needed; extra payables fund part of them. A net increase = cash outflow, added to the initial investment. At the end: that working capital is released and **returns as a cash inflow with no tax** in the terminal cash flow. Quick check: current assets +$22,000, current liabilities +$9,000 → NWC rises $13,000 (outflow at year 0).

Answer to Example 11.5 (Powell Corporation, tax 21%)

1. New machine installed cost = 380,000 + 20,000 = **$400,000**.
2. Old machine book value = 240,000 × (1 – 0.20 – 0.32 – 0.19) = 240,000 × 0.29 = **$69,600**.
3. Gain on sale = 280,000 – 69,600 = 210,400 → tax = 21% × 210,400 = **$44,184** → after-tax proceeds = 280,000 – 44,184 = **$235,816**.
4. Δ NWC = 35,000 – 18,000 = **$17,000** (outflow).
5. Initial investment = 400,000 – 235,816 + 17,000 = **$181,184**.

12 Operating cash flows and the terminal cash flow

Income statement format
Revenue
– Expenses (excluding depreciation and interest)
= EBITDA
– Depreciation
= EBIT
– Taxes (EBIT × tax rate)
= NOPAT
+ Depreciation (added back)
= **Operating cash flow (OCF)**

```
OCF = EBIT × (1 - T) + Depreciation
```

- Depreciation is not cash, but it lowers taxes — so it is added back.
- **Ignore interest**: how the project is financed is a separate decision (financing cost is already reflected in the cost of capital/WACC used for discounting).
- Replacement: incremental OCF = OCF with new machine – OCF with old machine.

Answer to Examples 11.6–11.7 (Powell, year 1, tax 21%)

	New machine	Old machine
Revenue – expenses	2,520,000 – 2,300,000 = 220,000	2,200,000 – 1,990,000 = 210,000
– Depreciation	80,000 (20% × 400,000)	28,800 (12% × 240,000)
= EBIT	140,000	181,200
– Tax 21%	29,400	38,052
= NOPAT	110,600	143,148
+ Depreciation	80,000	28,800
OCF	**190,600**	**171,948**

Incremental year-1 OCF = 190,600 – 171,948 = **$18,652**. Note: the new machine's EBIT is lower, but its large depreciation saves tax.

```
Terminal cash flow = After-tax proceeds from selling new asset -
After-tax proceeds from old asset (what we'd get without the
project) ± Working capital recovered
```

Same tax rule: compare each sale price with its book value at the end. Recovering working capital has no tax effect.

Answer to Example 11.8 (Powell, end of year 5)

New machine: sold $50,000, book value $20,000 → gain 30,000 → tax 6,300 → proceeds **$43,700**.

Old machine (if kept): sold $10,000, book value $0 → tax 2,100 → proceeds

$7,900.
Terminal cash flow = 43,700 – 7,900 + 17,000 = **$52,800**.

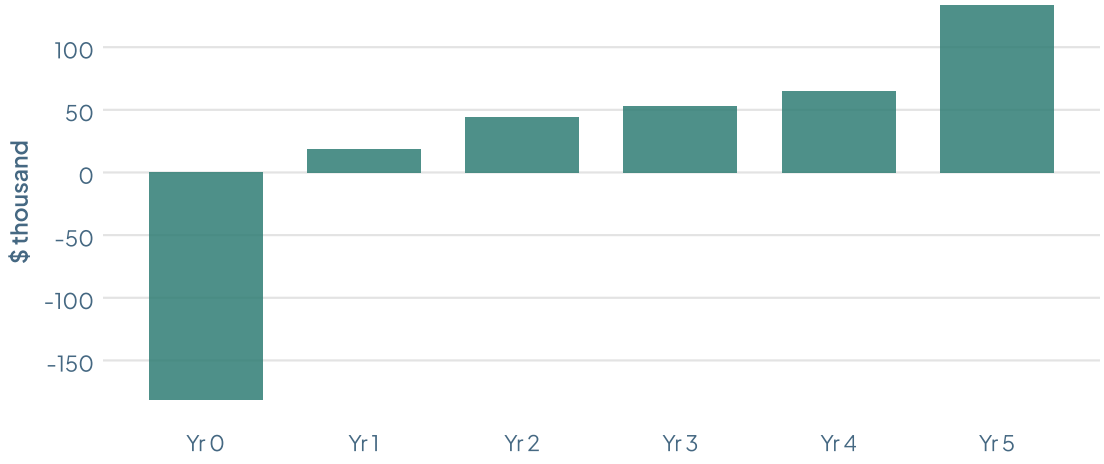
13 Putting it together: Powell's net cash flows and the decision

Initial investment + operating cash flows + terminal cash flow = the project's net cash flows:

Year	0	1	2	3	4	5
Powell	– $181,184	$18,652	$44,532	$52,940	$65,380	$133,980

Year 5 = OCF $81,180 + terminal $52,800. Then evaluate with the Chapter 10 tools (cost of capital 10%, illustrative):

Powell project net cash flows



Year 5 is large because it includes the $52,800 terminal cash flow. NPV at 10% = $40,197; IRR 16.3%.

NPV at 10%	IRR	Payback
$40,197 > 0 → accept	**16.3%** > 10% → accept	**≈ 4.0 years** (cumulative year 3 = 116,124; remaining 65,060 ÷ 65,380 ≈ 1.0)

Step 1 · Estimate the cash flows (Ch. 11): initial investment, OCF, terminal

→ Step 2 · Evaluate (Ch. 10): payback, NPV (+PI, EVA), IRR

→ Step 3 · Decide: NPV > 0 creates value; use NPV when methods conflict

! **Common mistakes**

(1) Putting interest into OCF — double counting, since it is already in the discount rate. (2) Forgetting tax on selling the old asset. (3) Forgetting to recover working capital in the final year. (4) Including sunk costs. (5) Using profit instead of cash.

Key takeaways

1. Capital budgeting = evaluating long-term investments (capital expenditures) that increase owners' wealth.
2. Process: proposal → review & analysis → decision → implementation → follow-up.
3. Payback = time to recover the investment; simple but ignores later cash flows and the time value of money.
4. NPV = $\Sigma\ CF_t \div (1+r)^t - CF_0$; accept if > 0. The best tool for deciding.
5. PI = PV of inflows ÷ investment; accept if > 1. EVA = cash flow – cost of capital × capital; PV of all EVAs = NPV.
6. IRR = the discount rate that makes NPV = 0; accept if IRR > cost of capital.
7. NPV and IRR always agree on accept/reject but can differ in ranking (reinvestment, timing, size). Bennett: NPV picks A, IRR picks B; lines cross near 10.7%.
8. Relevant cash flows: incremental, after tax, cash not profit. Ignore sunk costs, include opportunity costs.
9. Initial investment = installed cost – after-tax proceeds of old asset ± Δ working capital.
10. OCF = EBIT(1 – T) + depreciation; ignore interest. Terminal = new-asset proceeds – old-asset proceeds ± working capital recovered.

Case studies

Case & worked answers

Integrated practice: replacing a machine at PT Logam Prima

An illustrative practice example, not from the slides. PT Logam Prima (tax 22%, cost of capital 12%) is considering an **expansion** project: a IDR 900m machine + IDR 100m installation, straight-line depreciation over 5 years, no salvage value.

Extra revenue IDR 700m/year, extra cash expenses IDR 350m/year. Working capital rises IDR 80m at the start and is recovered in year 5. The machine is sold for IDR 150m at the end of year 5.

1. **Initial investment** = 1,000 + 80 = **IDR 1,080m** (expansion: no old asset).
2. **Annual OCF**: depreciation = 1,000 ÷ 5 = 200. EBIT = 700 – 350 – 200 = 150. Tax 22% = 33. NOPAT = 117. OCF = 117 + 200 = **IDR 317m**.
3. **Terminal cash flow**: book value 0 → gain 150 → tax 33 → proceeds 117; + working capital 80 → **IDR 197m**.
4. **NPV** = 317 × annuity factor (12%, 5) 3.6048 + 197 ÷ 1.12^5 – 1,080 = 1,142.7 + 111.8 – 1,080 ≈ **IDR 174.5m** → accept.
5. **PI** = 1,254.5 ÷ 1,080 ≈ 1.16. **Payback** = 1,080 ÷ 317 ≈ 3.4 years (excluding the terminal flow). IRR ≈ 17.9% > 12%.

Lesson: although a 3.4–year payback feels long, the positive NPV shows the project adds IDR 174.5m of value.

Worked examples

Problem 1 · Easy **Payback, NPV, PI and IRR for one project**

Project C: investment IDR 500m; year 1–4 cash flows: IDR 150, 200, 250, 100m. Cost of capital 12%, maximum payback 3 years. Compute payback, NPV, PI and estimate IRR. Accept or reject?

1. **Payback**: cumulative 150 → 350 → 600. Remaining after year 2 = 150; 150 ÷ 250 = 0.6 → **2.6 years** < 3 ✓.

2. **PV**: 150/1.12 = 133.93; 200/1.12² = 159.44; 250/1.12³ = 177.95; 100/1.12⁴ = 63.55 → total 534.86.

```
NPV = 534.86 - 500 = 34.86
```

3.

```
PI = 534.86 ÷ 500 ≈ 1.07
```

4. **IRR** by trial: at 15% NPV = +3.2; at 16% NPV = −6.7 → IRR ≈ 15 + 3.2 ÷ (3.2 + 6.7) ≈ **15.3%** > 12%.

5. All criteria agree: **accept**.

Answer: Payback 2.6 yrs; NPV IDR 34.9m; PI 1.07; IRR ≈ 15.3% → accept.

Problem 2 Easy **NPV and EVA of a perpetuity project**

An IDR 2,000m project produces IDR 260m a year forever. Cost of capital 11%. Compute NPV and annual EVA.

1. `NPV = CF ÷ r - Investment = 260 ÷ 0.11 - 2,000 = 2,363.6 - 2,000 = 363.6`

2. `EVA = 260 - 0.11 × 2,000 = 260 - 220 = 40`

3. Check: PV of EVA = 40 ÷ 0.11 = 363.6 = NPV ✓.

Answer: NPV ≈ IDR 363.6m; EVA IDR 40m/year.

Problem 3 Medium **Initial investment of a replacement project**

New machine IDR 600m + IDR 40m installation. The old machine was bought 4 years ago for IDR 300m, depreciated by MACRS 20% + 32% + 19% + 12%, and can be sold for IDR 120m. Tax 22%. Working capital rises IDR 30m. Compute the initial investment.

1. Installed cost = 600 + 40 = **640**.
2. Old book value = 300 × (1 − 0.83) = **51**.
3. Gain = 120 − 51 = 69 → tax = 22% × 69 = 15.18 → after-tax proceeds = 120 − 15.18 = **104.82**.
4.
```
Initial investment = 640 - 104.82 + 30 = 565.18
```

Answer: Initial investment ≈ IDR 565.2m.

Problem 4 Medium **Incremental OCF**

Continued. Year 1 with the new machine: revenue 1,000, cash expenses 700, depreciation 128 (20% × 640). With the old machine: revenue 900, cash expenses 680, depreciation 36 (12% × 300). Tax 22%. Compute year-1 incremental OCF.

1. ```
   OCF = EBIT × (1 - T) + Depreciation
   ```
2. New: EBIT = 1,000 − 700 − 128 = 172 → 172 × 0.78 + 128 = **262.16**.
3. Old: EBIT = 900 − 680 − 36 = 184 → 184 × 0.78 + 36 = **179.52**.
4. Incremental = 262.16 − 179.52 = **82.64**. (Interest ignored.)

Answer: Year-1 incremental OCF ≈ IDR 82.6m.
   ```

Problem 5 Medium **Terminal cash flow**

At project end: the new machine sells for IDR 90m (book value IDR 35m); the old machine — if kept — would sell for IDR 15m (book value 0). Working capital of IDR 30m is recovered. Tax 22%. Compute the terminal cash flow.

1. New: gain 55 → tax 12.1 → proceeds 90 − 12.1 = **77.9**.
2. Old: gain 15 → tax 3.3 → proceeds 15 − 3.3 = **11.7**.
3.
```
Terminal = 77.9 - 11.7 + 30 = 96.2
```
Working-capital recovery is not taxed.

Answer: Terminal cash flow ≈ IDR 96.2m.

Practice questions

Try answering first. The answer key is below each question.

1. An example of a capital expenditure is…

A. Paying monthly salaries
B. Buying raw materials
C. Building a new factory
D. Paying the electricity bill

Answer: C. A factory's benefits last over a year; it is large and hard to reverse.

2. Two machines for the same job — choosing one rules out the other. These projects are…

A. Independent
B. Mutually exclusive
C. Capital rationing
D. Accept–reject

Answer: B. Competing projects are mutually exclusive.

3. Bennett's Project A payback (investment 420,000, 140,000 a year) is…

A. 2.5 years
B. 2.75 years
C. 3.0 years
D. 3.5 years

Answer: C. 420,000 ÷ 140,000 = 3.0 years — over the 2.75 limit, so rejected on payback.

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4. The main weakness of the payback period is…

A. Too complex to compute
B. Ignores cash flows after payback and the time value of money
C. Always picks the largest project
D. Requires the cost of capital

Answer: B. Payback stops counting once the investment is recovered and does not discount.

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5. Under NPV a project is accepted if…

A. NPV > 0
B. NPV > cost of capital
C. NPV > 1
D. NPV = IRR

Answer: A. A positive NPV means the project earns more than investors require.

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6. PV of inflows 559,244, investment 450,000. The PI is…

A. 0.80
B. 1.10
C. 1.24
D. 1.26

Answer: C. 559,244 ÷ 450,000 ≈ 1.24 (Project B).

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7. A $1m project earns $120,000 a year forever, cost of capital 10%. Its annual EVA is…

A. $12,000
B. $20,000
C. $120,000
D. $200,000

Answer: B. 120,000 – 10% × 1,000,000 = 20,000. Its PV 20,000 ÷ 0.1 = 200,000 = NPV.

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8. IRR is the discount rate that makes…

A. Payback = 0
B. NPV = 0
C. PI = 0
D. EVA = cost of capital

Answer: B. IRR = the project's annual return; at that rate NPV is exactly zero.

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9. Bennett: NPV picks A, IRR picks B. Cost of capital 10%. The right decision is…

A. B, higher IRR
B. A, higher NPV
C. Reject both
D. B, faster payback

Answer: B. When they conflict, decide with NPV: it measures wealth added and has a realistic reinvestment assumption.

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10. Option 1: invest $2, get $3. Option 2: invest $1,000, get $1,100. Which adds more wealth?

A. Option 1 (IRR 50%)
B. Option 2 (adds $100)
C. The same
D. Cannot tell

Answer: B. IRR ignores size. Option 2 adds $100 vs $1.

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11. NPV assumes early cash is reinvested at…

A. The project's IRR
B. The cost of capital
C. Zero percent
D. The deposit rate

Answer: B. The cost-of-capital assumption is more realistic than IRR's.

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12. Market research already paid for last year, in project analysis, is…

A. Included as a cost
B. Ignored as a sunk cost
C. Included as an opportunity cost
D. Depreciated

Answer: B. A sunk cost does not change with the decision, so it is irrelevant.

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13. An asset with $48,000 book value sells for $30,000, tax 21%. The tax effect is…

A. Pay $6,300 tax
B. Save $3,780 tax
C. No tax
D. Pay $3,780 tax

Answer: B. Loss of 18,000 × 21% = 3,780 tax saving (cash inflow).

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14. Current assets rise $35,000, current liabilities rise $18,000. The effect on the initial investment is…

A. $17,000 outflow
B. $17,000 inflow
C. $53,000 outflow
D. None

Answer: A. Δ NWC = 35,000 – 18,000 = 17,000, added to the initial investment and recovered at the end.

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15. When computing project operating cash flow, interest expense is…

A. Deducted before tax
B. Ignored, since financing is a separate decision
C. Added back like depreciation
D. Deducted after tax

Answer: B. Financing cost is already in the discount rate; putting it in OCF double counts.

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16. EBIT 140,000, tax 21%, depreciation 80,000. OCF is…

A. 110,600
B. 140,000
C. 190,600
D. 220,000

Answer: C. 140,000 × 0.79 + 80,000 = 110,600 + 80,000 = 190,600.

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17. Recovering working capital at the end of a project is…

A. Fully taxed
B. A cash inflow with no tax
C. Ignored
D. A cash outflow

Answer: B. Working capital simply comes back; there is no gain, so no tax.

Sources

- P3MD Week 9 MK2 — Capital Budgeting Techniques and Cash Flows

- Zutter & Smart — Principles of Managerial Finance, Ch. 10 & 11 (Pearson)